

DAY 04

LECTURE · 90 MINUTES

Out to the Edge.

SUBTITLE

When the cloud is too far
away

TODAY

Latency · IoT & 5G

THEN

Edge compute · Edge AI

TOMORROW

The capstone

A QUESTION THE CLOUD CANNOT ANSWER

A self-driving car sees a child step out. **Should it brake?** There is no time to ask a data centre a thousand kilometres away.

FOR THREE DAYS WE SENT EVERYTHING TO THE CLOUD. TODAY WE
LEARN WHEN, AND WHY, TO KEEP IT CLOSE.

COMPUTING IS LEAVING THE BUILDING

The edge is the cell tower, the shop, the car, **the phone in your hand.**

Five questions, **one step closer.**

TODAY'S LECTURE

§ 01

Why isn't the cloud enough?

The tyranny of distance, and the things even the cloud cannot beat.

§ 02

What is pushing computing outward?

Billions of devices, the internet of things, and 5G.

§ 03

How does the edge actually run?

From caching content to running your code in hundreds of cities.

§ 04

Can the intelligence move too?

Edge AI, and the model already running on your phone.

§ 05

So what lives where?

Device, edge and cloud together. The question your capstone answers.

— SECTION ONE

01

The tyranny of **distance.**

The cloud is powerful, but it is **far**. And for some things, far is the one thing you cannot afford.

• THE CAR • 08:14:00 • A CHILD STEPS INTO THE ROAD

REMEMBER DAY 2

Light is fast, but it is not instant. A round trip to a distant region is **tens or hundreds of milliseconds.**

FOR WATCHING A FILM, YOU NEVER NOTICE. FOR A CAR, A ROBOT ARM, OR A SURGEON'S INSTRUMENT, THAT DELAY IS THE DIFFERENCE BETWEEN SAFE AND DANGEROUS.

Three things the cloud **cannot** fix.

NO MATTER HOW BIG THE DATA CENTRE

01

Delay

Distance costs time, and physics sets the floor. The only way to respond faster is to be closer.

02

The cost of shipping data

Sending every camera frame and sensor reading across the world is slow and expensive. Most of it is not worth the trip.

03

Losing the connection

Tunnels, ships, factories, remote roads. The cloud is useless the moment the signal drops. The edge keeps working.

A SIMPLE DEFINITION

Computing, **close to the action.**

The edge means doing the work **near where the data is born**, instead of shipping it all to a faraway cloud.

It is not one place. It is a spectrum: the chip in your phone, a box in the shop, a small data centre at the cell tower. **Anywhere closer than the cloud.**

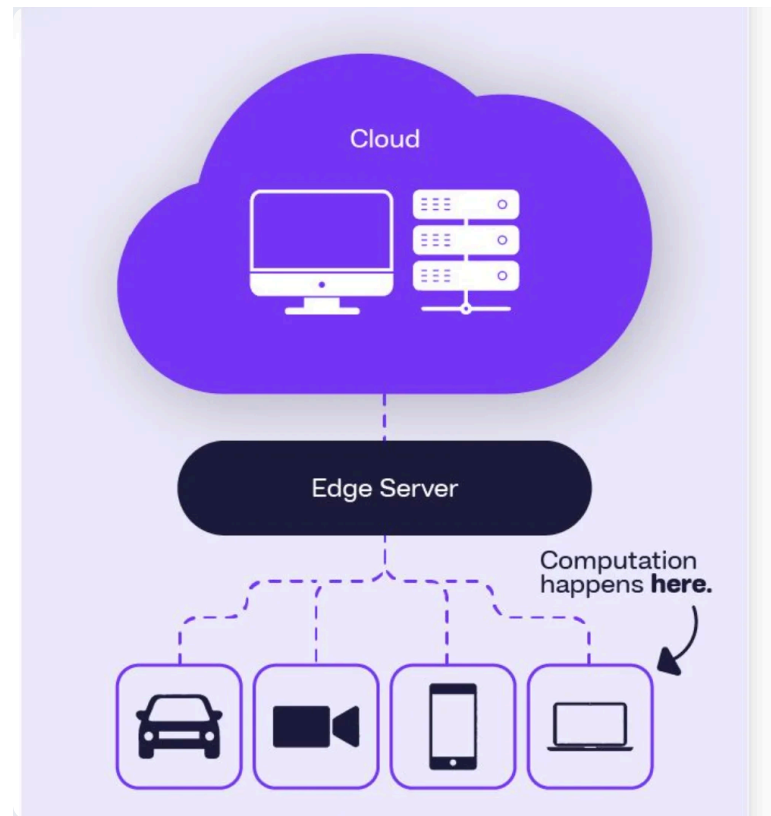


FIG. WORK HAPPENS NEARBY

Same request, **two distances.**

WATCH THE ROUND TRIP COST YOU

Brake the car

AR overlay

Voice reply

At the edge ON THE SPOT

3 ms

RESPONDED

To the cloud and back ROUND TRIP

63 ms

A self-driving car deciding whether to brake cannot wait for a round trip to a data centre. At **140 ms**, the car has already travelled metres. The decision has to happen **on the car**.

Not edge **or** cloud. A **continuum.**

FROM YOUR HAND TO THE HYPERSCALER

Closest

The device

Your phone, watch or car. Instant, private, but small and battery-bound.

Near

The local edge

A box in the building or at the cell tower. Milliseconds away, just for you.

Regional

The edge city

A small data centre in your city, shared by everyone nearby.

Farthest

The cloud

The giant region. Endless power and storage, but the slowest to reach.

The skill is not picking one. It is putting each job at the **right point on this line.**

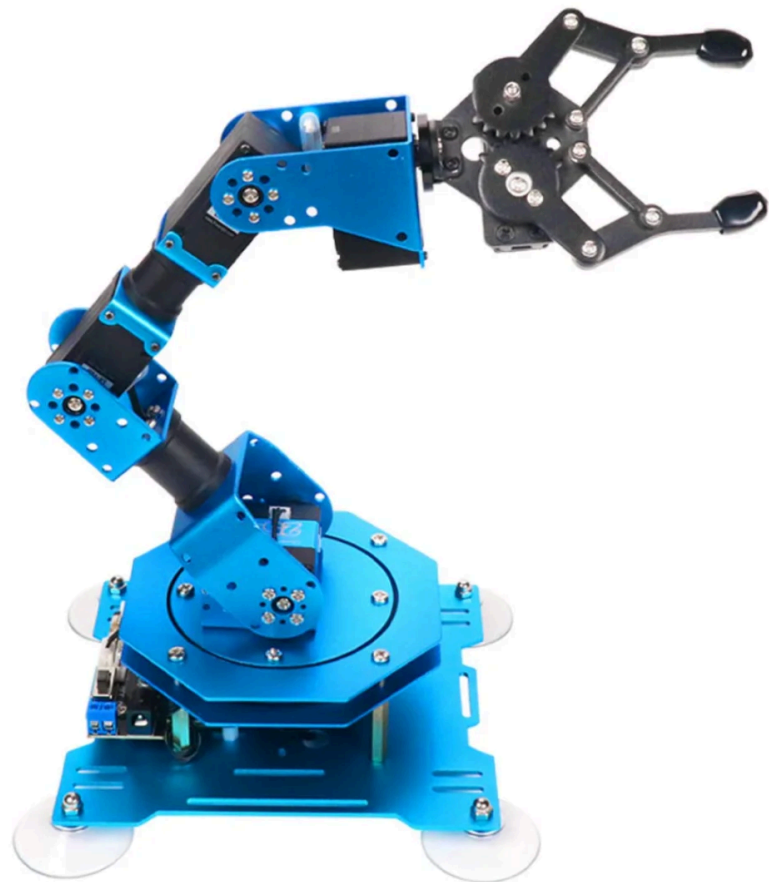
• STORY • THE HUMAN NERVE

Your own body chose the edge long ago.

Touch something hot and your hand pulls back before your brain even knows. The signal never travels all the way up; a reflex is handled **right at the spinal cord**.

Evolution learned that for danger, the round trip to headquarters is **too slow**. Self-driving cars and robots are rediscovering exactly the same lesson.

The lesson: keep the reflexes local, and save the deep thinking for the cloud.



Race your own **nervous** **system.**

START THE TEST

CLICK THE PAD THE INSTANT IT TURNS AMBER

WAITING

Your reflex		— ms
Cloud round trip		140 ms
The edge		5 ms

— SECTION TWO

02

A world full of **sensors.**

The edge did not arrive by choice. It was **pushed** into being by billions of small, chattering devices.

• THE CAR • ONE DAY OF DRIVING • TERABYTES BORN ON BOARD

CONNECTED THINGS ON EARTH

tens of

billions of devices are now online: doorbells, watches, cars, factory machines, farm sensors, city lights. They already **outnumber people several times over**, and almost none of them is a computer you would recognise.

THE INTERNET OF THINGS

Everyday objects that **sense** and **speak**.

For decades, only computers and phones were online. **IoT** is the wave of ordinary objects joining them: a thermostat, a pallet, a heart monitor, a streetlight.

Each one quietly **senses** the world and **reports back**. Multiply that by billions and you have a planet covered in eyes and ears.

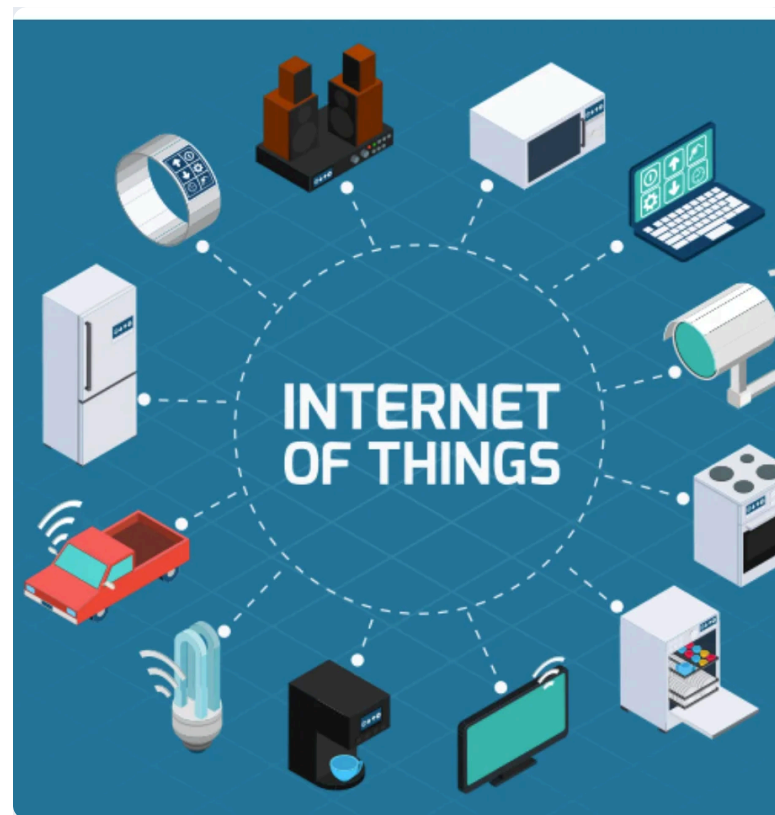


FIG. ORDINARY THINGS, ONLINE

WHY ALL OF THIS FORCES THE EDGE

A billion sensors make **an ocean of data.** Almost none of it is worth keeping.

SHIPPING EVERY READING TO THE CLOUD WOULD DROWN THE NETWORK AND THE BUDGET. FAR BETTER TO LOOK AT IT WHERE IT IS MADE, AND SEND ON ONLY WHAT MATTERS.

Drink the ocean, **send a cup.**

WHAT THE EDGE FILTERS OUT BEFORE THE CLOUD EVER SEES IT

MADE AT THE EDGE

2.0_{GB}

every minute, from one machine's sensors



FILTERED **99.8%** ON THE SPOT

SENT TO THE CLOUD

4_{MB}

just the summary worth keeping

MORE THAN FASTER PHONES

5G is built for **the edge**.

People think 5G means quicker downloads. Its quieter gifts matter more: **very low delay**, and the ability to connect a **huge density** of devices at once.

It even lets operators place small data centres **right at the tower**. The network itself becomes a layer of the edge.



FIG. COMPUTE AT THE TOWER

A backhoe finds your only cable.

CUT THE FIBRE · 02:13

ONE REMOTE FACTORY, TWO DESIGNS, THE SAME BAD MORNING

DESIGN A · ALL IN THE
CLOUD

● FIBRE LINK · UP

Sensors reporting OK

Robot arms OK

Quality checks OK

Safety alerts OK

RUNNING

DESIGN B · EDGE BOX ON
SITE

● FIBRE LINK · UP

Sensors reporting OK

Robot arms OK

Quality checks OK

Safety alerts OK

Buffered locally · 0 MB

RUNNING

• STORY • THE DATA ON WHEELS

A single modern car is a **data centre on wheels.**

Cameras, radar and lidar on an advanced vehicle can generate **terabytes of data in a single day of driving.** No mobile network on Earth could stream all of that to the cloud in real time.

So the car decides for itself, in the moment, and sends home only what is worth learning from later. **It has to be its own edge.**

The lesson: when a device makes more data than it can ever send, the computing must move to the data.



— SECTION THREE

03

The edge that **runs your code.**

You have already met the edge without knowing it. **Today** it stopped just storing things, and started doing them.

• THE CAR • THE CITY AROUND IT LEARNS TO THINK

A CALLBACK TO DAY 2

The CDN that kept a copy of the website near you? **That was the edge all along.** It just held still pictures.

FOR YEARS THE EDGE COULD ONLY CACHE AND SERVE FILES. THE BREAKTHROUGH WAS TEACHING THOSE SAME NEARBY MACHINES TO RUN ACTUAL PROGRAMS.

THE LEAP

Your code, in **hundreds of cities.**

Modern edge platforms let you upload a small piece of code and run it in **every one of their cities at once.** A request is handled by whichever is nearest the user.

It is yesterday's serverless idea, scattered across the planet. **This is exactly what today's lab will do.**

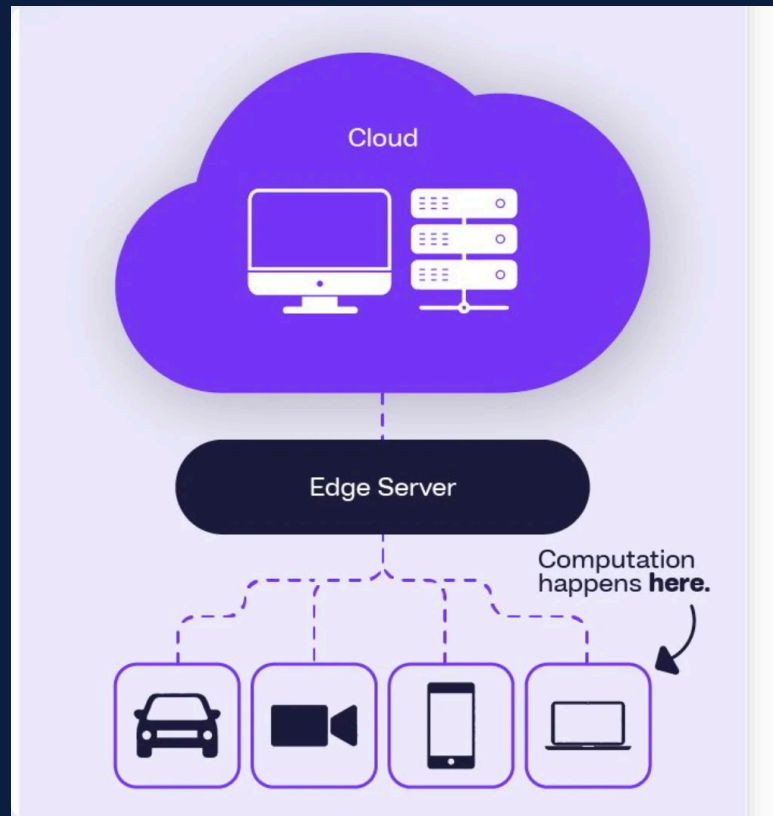


FIG. ONE FUNCTION, EVERYWHERE

What belongs **at the edge?**

SMALL, FAST JOBS THAT EVERY USER NEEDS

01

Checking who you are

Logins and permissions, settled at the nearest city before a request travels any further.

02

Tailoring the page

The right language, prices and layout for your region, assembled close to you.

03

Resizing and reshaping content

Shrinking an image to fit your screen, on the spot, instead of storing a hundred versions.

04

Blocking the bad traffic

Stopping bots and attacks at the door, far from the precious origin servers.

One upload, **the whole planet.**

A SINGLE EDGE FUNCTION, LIVE EVERYWHERE AT ONCE

0

CITIES LIVE, IN SECONDS

► Deploy again

Jeddah

5 ms

Dubai

6 ms

Cairo

7 ms

Frankfurt

5 ms

London

4 ms

Mumbai

7 ms

Singapore

6 ms

Tokyo

5 ms

Sydney

8 ms

Lagos

11 ms

New York

4 ms

São Paulo

9 ms

THE EDGE AS A BORDER GUARD

Keep the data **at home**.

Yesterday we saw that some data is not allowed to leave its country. The edge turns that rule into an advantage: process the data **in the city where it was made**, and only a harmless summary ever crosses a border.

Sensitive records stay put, the law is satisfied, and the service still feels **instant**.



FIG. PROCESS LOCALLY, SHARE LITTLE

• STORY • THE LEVEL PLAYING FIELD

Two students can now serve the whole world.

Not long ago, being fast everywhere meant renting servers on every continent: a job for a large company with a large budget.

Today a student can write one small function, type a single deploy command, and have it running **in hundreds of cities a minute later**, fast for everyone on Earth. The same reach Netflix built over years.

The lesson: the edge did not just move computing closer. It put global scale within reach of **anyone**.



— SECTION FOUR

04

Intelligence at **the edge.**

First we moved the storage out. Then the computing. Now the most surprising thing of all is moving too: **the AI.**

• THE CAR • THE BRAIN CLIMBS ABOARD

A QUIET REVOLUTION IN YOUR POCKET

We picture AI as vast machines in a distant data centre. Yet some of it now runs **entirely on the phone in your hand.**

MODELS HAVE BEEN SHRUNK AND CHIPS HAVE BEEN SHARPENED TO THE POINT WHERE REAL INTELLIGENCE FITS IN A DEVICE THAT RUNS ON A BATTERY.

Why keep the AI **close**?

THE SAME FOUR REASONS, ONE MORE TIME

01

Speed

An answer with no round trip feels instant. Unlocking your phone cannot wait on a network.

02

Privacy

Your face, your voice and your photos never have to leave the device to be understood.

03

It works with no signal

Live translation on a plane, dictation in a tunnel. No connection required.

04

Cost

Running on hardware the user already owns costs the company almost nothing per use.

PICK A TASK

01 Face unlock

02 Live translate

03 Enhance a photo

04 Voice assistant

ON YOUR DEVICE

BEST HERE

LATENCY **Instant**PRIVACY **Never leaves**OFFLINE **Always works**

IN THE CLOUD

WRONG TOOL

LATENCY **Round trip**PRIVACY **Your face uploaded**OFFLINE **Useless offline**

WHY WE STILL NEED THE CLOUD

A phone is not a data centre. On-device models must be **small**, and small models cannot do everything.

THE LARGEST, SMARTEST MODELS STILL LIVE IN THE CLOUD. THE ART IS SPLITTING THE WORK: THE QUICK, PRIVATE, PERSONAL PART ON THE DEVICE, THE HEAVY THINKING IN THE CLOUD.

● STORY • YOU ALREADY USE IT

You have trusted edge AI **a thousand times.**

Every time your phone recognises your face to unlock, it runs a neural network **on the device, in a fraction of a second.** The map of your face never leaves the phone, and never touches the internet.

The same is true of live captions, dictation, and the quiet magic that sharpens every photo you take. **All of it, on the edge, in your pocket.**

The lesson: the future we are describing is not coming. For the small, personal moments, it already arrived.



— SECTION FIVE

05

Putting it **together.**

Device, edge and cloud are not rivals. They are **a team**, and the whole game is knowing who plays which part.

• THE CAR • THREE MILLISECONDS LATER • IT HAS ALREADY BRAKED

One job, **three** homes.

HOW A SMART DOORBELL REALLY WORKS

Device

Sees motion

The camera detects movement and decides, on the spot, that something moved.

Edge

Is it a person?

A nearby model checks in milliseconds whether it is a human, a cat or a passing car.

Cloud

Who is it?

Only if needed, the big model matches a known face and stores the clip for later.

Back

Your phone buzzes

A notification arrives, and a quick on-device model wrote the caption.

Each layer does only what it is **best placed to do**. That cooperation is the *real* architecture of modern computing.

THE ONE QUESTION THAT TIES THE WEEK TOGETHER

Every system you will ever design comes down to one decision: **what lives where?**

REFLEXES ON THE DEVICE. FAST, SHARED, LOCAL WORK AT THE EDGE.
HEAVY LIFTING AND LONG MEMORY IN THE CLOUD. GET THAT SPLIT
RIGHT AND EVERYTHING ELSE FOLLOWS.

Six jobs. **You** **place them.**

JOB 1 OF 6

Decide, right now: brake or not?

DEVICE, EDGE OR CLOUD — A WARM-UP FOR TOMORROW'S CAPSTONE

The device

ON THE THING ITSELF

The edge

A NEARBY BOX OR CITY

The cloud

THE BIG REGION, FAR AWAY



The honest **trade-off.**

NEITHER SIDE WINS OUTRIGHT

THE CLOUD

Vast, **but far.**

Endless power, storage and the biggest models.
Simple to manage in one place. The price is distance,
and the delay that comes with it.

THE EDGE

Instant, **but small.**

Close, fast, private, and works offline. The price is
limited power, and the headache of running things in
thousands of places at once.

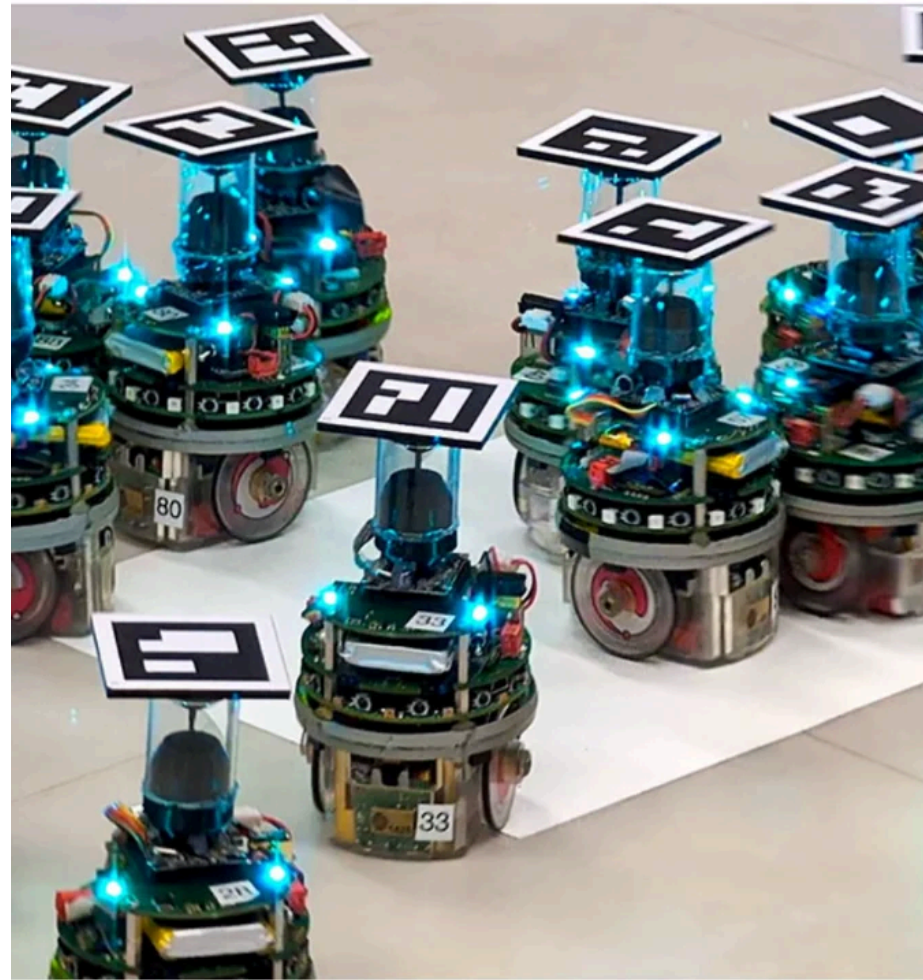
• STORY • 2016

A million cameras became **one weapon.**

In 2016, attackers quietly took over hundreds of thousands of cheap webcams and doorbells, devices whose owners never changed the default password. Together, this army of gadgets was pointed at the internet's plumbing and **knocked large parts of it offline.**

Every device at the edge is also a tiny computer that someone else might break into. **Billions of edges mean billions of doors.**

The lesson: pushing computing everywhere is powerful, and it spreads the risk everywhere too. Convenience and security pull in opposite directions.



Tomorrow, **you decide.**

THE CAPSTONE, IN ONE SENTENCE

A A streaming service

Where do you cache, transcode and store? How do you stay fast across continents?

B A hospital network

Which data may leave the building? What must keep working when the link drops?

C A smart-city traffic system

What decides at the junction, and what is learned later, centrally?

+ On-prem · Cloud · Edge

For your scenario, decide what lives where, then defend it. That is the whole week, applied.

— THE REFRAME

For thirty years computing rushed **into the cloud**.
Now it is rushing back out, all the way to **your hand**.

You have seen the building, the journey, the rental and the edge. **Tomorrow**
you put it all together, and design real infrastructure for a real scenario. The
capstone.

LAB NOW • DEPLOY TO THE EDGE, THEN MEASURE YOUR RESPONSE TIME FROM
AROUND THE WORLD

DAY 05 → THE
CAPSTONE